

Account aggregator ecosystem: A step towards revolutionising digital lending in India

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Abstract An essential contributor to a nation's economic success is providing individuals with access to established financial systems. India, through innovative use of digital finance infrastructure, aka India Stack, provided as a public good, removes obstacles in offering financial products to the unbanked and underbanked. The three pillars of the India Stack are: the identity rail (Aadhaar), the payment rail (universal payment interface) and the data-sharing rail of account aggregators (AAs). The next phase in the India Stack is the Open Credit Enablement Network (OCEN), which promises to transform the digital lending industry by shifting the business model from asset-based to cash-flow-based lending. OCEN is a standard set of application programming interfaces (APIs) that pull data based on customer consent from financial information providers and offers it to financial information users to decide on credit requests. AA manages consent on behalf of the customer using data empowerment and protection architecture (DEPA). The rest of the paper explains the interoperability and portability features of the AA ecosystem, the pricing model that the AA can adopt for a sustainable business model, describes the current state of the AA ecosystem in terms of consents provided and fulfilled and concludes with the use cases of OCEN in credit extension and continuous loan monitoring.

KEYWORDS: account aggregator (AA), India Stack, Open Credit Enablement Network (OCEN), open banking, digital lending, data empowerment and protection architecture (DEPA)

INTRODUCTION

India has a massive credit deficit. According to one estimate, a credit gap of approximately US\$330bn is attributed to traditional lenders' inability to extend credit to 'new to banking' and 'new to credit' customers. Conventional lending is asset based and requires collateral to obtain credit. The process of credit underwriting is cumbersome, requiring the borrower to provide bank statements, insurance policies, mutual fund holdings and tax papers, in physical form, besides delaying credit decisions and increasing the likelihood of errors and inefficiency. The account aggregator (AA) platform through the Open Credit Enablement Network (OCEN) utilising India Stack with three rails for identity, payment and data sharing aims to alleviate lenders' difficulties when extending credit. OCEN seeks to replace asset-based lending with cash-flow-based lending. Small ticket digital transactions and QR code-based payments are increasing small merchants' banking transactions, allowing a shift to cash-flow-based lending.¹ These financial transactions or statement information are crucial for credit decision making in cash flow-based lending.

OCEN through standardised APIs and customer consent enables sharing of consumers' financial transaction data with all the regulated entities in a safe and secured manner, with data being encrypted end to end. Availability of the consumers' financial transaction data may allow financial entities to create new propositions that meet the latent demand of the consumers, giving them tangible benefits. Brett King, in his book, *Bank 4.0*,² stated that the availability of personal financial data through standardised APIs and open banking protocols will transform the banking industry from 'closed' banking to 'open' banking and in the process give consumers highly personalised services and 'me-centred' experience wherever, however and whenever the customer wants.³

The remainder of the paper describes the obstacles to credit expansion, the proposal

to overcome the barriers to credit expansion and the proposal's explanation of the India Stack and its identity rail, payment rail and data-sharing rail. The paper describes how AA utilise OCEN to address the challenges lenders and borrowers face during the credit process. It describes the participants in the new credit ecosystem facilitated by OCEN. It examines the institutional and technical architecture to explain data interoperability and portability among the AA ecosystem players. It discusses how the participants in the OCEN ecosystem interact and how this integration differs from the open banking ecosystem of the United Kingdom made possible by PSD2 (Payment Services Directive). The paper also touches on the revenue model that the AA can adopt. Finally, the use cases are provided to describe the role of OCEN in financial inclusion.

CHALLENGES TO CREDIT EXPANSION

There are three dimensions to the challenges to credit expansion: (1) challenges for the lenders, (2) challenges for borrowers and (3) information technology challenges.

Challenges for the lenders

The challenges that lenders face are (a) identification of creditworthy borrowers — lenders' cost of acquiring a borrower is exceptionally high, particularly in situations where borrowers desire low-value, short-term loans; (b) lack of reliable information to assess the creditworthiness and repayment capacity of the borrowers; and (c) cash flow collection: it can be challenging for lenders to collect money.

Challenges for the borrowers

The challenges that borrowers confront are banks and non-banking financial companies (NBFCs) needing physical collateral to lend money. (a) Lenders continue to rely on a manual decisioning and underwriting

process, as opposed to an automated one, which causes delays in disbursement. In the absence of collateral, processing a loan is expensive because it requires borrowers to meet with underwriting officers in a branch.

(b) Low liquidity prevents microfinance institutions from lending to needy borrowers. Both their interest rates and transaction fees are incredibly high. (c) Loan sharks are people who operate outside of the formal financial system and engage in predatory lending. They give high interest loans.

Lenders face obstacles to issuing loans, and borrowers face obstacles to obtaining credit. Borrowers, particularly micro, small and medium-sized enterprises (MSME), have no access to credit when required, nor do they receive sufficient credit. The credit delivery rail for the flow of capital is broken for the MSME sector, despite a 20–25tn-rupee credit gap.⁴

Information technology challenges

Digital lending has been proposed to alleviate the difficulties faced by retail consumers and small and medium banking (SMB) borrowers in gaining timely access to credit. Credit market participants who wish to enter this digital lending space, however, are confronted with various technical challenges: (a) Integration of non-standard and non-standard-based software systems. Market participants must integrate with financial institutions like banks and NBFCs for credit evaluation, disbursement and payment collection. In the absence of standards, market participants must integrate with these entities uniquely. In certain instances, marketplaces are required to connect with these entities via spreadsheet manually. (b) Extended credit decisioning cycles because banks and NBFCs continue to use manual credit appraisal systems. (c) Inability to tailor products to borrowers' requirements. Banks and NBFCs offer one-size-fits-all credit products without considering the borrower's particular needs.

Products are not based on the borrower's personas, and credit products are not sized, priced and timed optimally.

INDIA STACK

India intends to address the difficulties associated with credit development by embracing paperless, presence-less and cashless service delivery through a technology stack known as the India Stack. These are autonomous yet interconnected systems. They are called stack because they are modular and may be stacked in any manner to construct a digital infrastructure. They are also known as 'platforms', enabling users to produce and consume value.⁵ India has chosen to construct the technology stack that is the foundation of its digital infrastructure from scratch. It has opted to offer digital infrastructure as a public good to prevent technology companies from controlling the platform. In his analysis, Nilekani observed that 'Aadhaar demonstrates India's commitment to the principle that digital infrastructure should be a public good'.⁶ The digital infrastructure is open and interoperable, allowing public and private sector partners to develop technological innovation and products that leverage this infrastructure within the regulatory framework.

The India Stack consists of three base layers, or rails.⁷ As seen in Figure 1, several specialised single-purpose utilities and platforms have been developed on top of Aadhaar including e-KYC, Digi locker and e-Sign, as well as payment systems (UPI, IMPS). (1) Aadhaar creates an identity layer that is presence-less and paperless, giving each resident a distinct identity. Aadhaar is provided without a clear purpose or entitlement, and its sole goal is to enable an individual to be identified uniquely. (2) A cashless payments layer enables anyone to pay anyone else using a universal payment interface (UPI) efficiently and cost-effectively. (3) Finally, a data layer

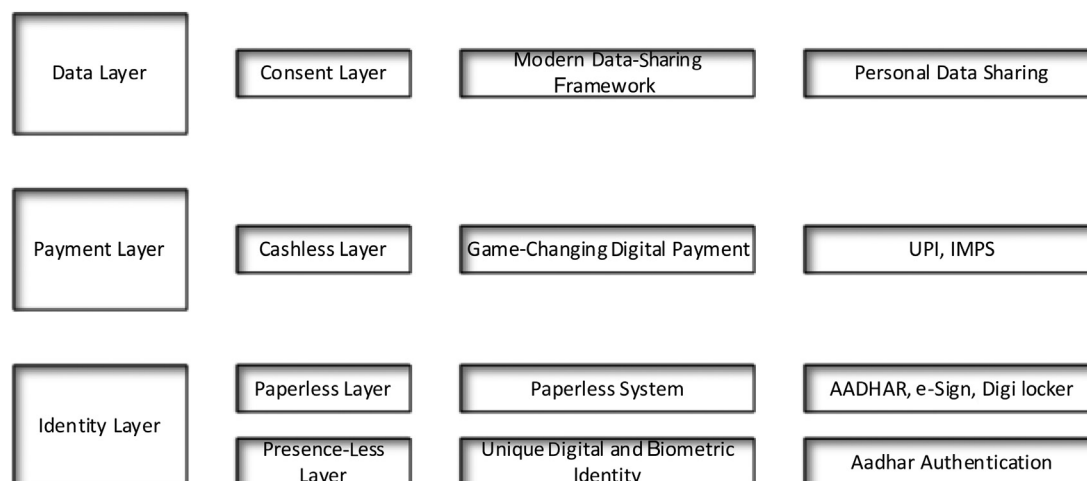


Figure 1 India Stack

supported by consent architecture enables secure personal data sharing without invading privacy, such as Digi locker.

India Stack induced a paradigm change in the financial services industry and revolutionised how underbanked and unbanked persons acquire financial products.⁸ OCEN, which is the next chapter in the India Stack, is envisioned as a data-sharing rail to ensure that credit distribution can be democratised to promote financial inclusion and increase the availability of credit products for consumers who are New to Bank (NTB) or New to Credit (NTC).

NEW CREDIT ECOSYSTEM – OPEN CREDIT ENABLEMENT NETWORK

What is OCEN?

OCEN is a framework of APIs that enables integration between loan service providers (LSPs), financial institutions and AA, thereby creating a new credit ecosystem. OCEN is an open credit protocol that will facilitate the creation of customised credit products for businesses and individuals while leveraging the Open Payments protocol of UPI for disbursing and collecting payments. OCEN through open credit protocol and standardised APIs will help banks leverage

consumer data buried in proprietary IT systems of different financial and tax entities. Access to consumer data will transform the banking model of the future into some form of marketplace banking where the traditional banking business model will be transformed into a data-intensive, platform-based marketplace, where several financial entities will continuously compete to offer customers tailored, good-value products.⁹

Who are the players in OCEN ecosystem?

The principal participants in this new credit ecosystem are LSPs, AA, lenders (users of financial information) and financial information providers.

Loan Service Providers (LSPs): They serve as borrowers' agents and offer protection to borrowers so that the issue of mis-selling is avoided. LSP differs from the DSA (Direct Sourcing Agent) model, in which the agent works for the financial institution. By partnering with banks and non-banking financial institutions, LSP helps borrowers obtain personalised credit from formal channels at a lower interest rate.

Financial Information Providers (FIPs): These are entities that provide financial data generated by a consumer to a requester of the data with the consumer's consent.

Banks, mutual fund companies, insurance companies, telecommunications companies and the Goods and Services Tax Network (GSTN) are FIPs.

Financial information Users (FIUs) use financial data to provide products and services, such as credit, personal finance management, insurers and wealth managers.

Account aggregators (AAs): Account aggregator is a service that collects financial information from various financial data providers such as banks, GSTN and telecom providers and consolidates, organises and presents it to the end customer and the authorised FIU via a consent-based, simple and secure digital access service.¹⁰

What is an account aggregator?

Advances in technology and the evolution of the digital landscape, fuelled by the always-on interconnected web, have enabled regulatory interventions that have the potential to drive truly revolutionary change in financial services. In 2016, the Reserve Bank of India (RBI) made an intervention and authorised a new category of NBFCs to serve as AA, entities responsible for managing user consent for financial data sharing. AA is a data access trustee for financial data created by RBI, Securities and Exchange Board of India (SEBI), Insurance Regulatory and Development Authority of India (IRDAI) and Pension Fund Regulatory and Development Authority (PFRDA) via the Financial Stability and Development Council. The AA ecosystem only lets people share data about assets, such as savings accounts, checking accounts, fixed deposits, mutual funds, pension fund accounts and insurance policy data. Access to transactional data is buried within the siloed proprietary IT systems of the banks and other financial entities.

AA offer FIU services that include gathering, compiling, consolidating and sharing financial data for an individual. AA provides a place for FIUs and FIPs to

share and use information. AA facilitates financial data flow between various regulated financial institutions such as banks, securities firms and pension funds. Access to the data is provided to the regulated entity for a limited time and for a specific purpose and, that too, with the customer's full knowledge and consent, giving the customer full control over the distribution of their data.

What are the main responsibilities of account aggregator and its operating model?

According to the RBI's master directions¹¹ in 2016, and updated in December 2022, the essential duties and responsibilities of an AA are (1) to record explicit consent of a customer to share data; (2) to provide services to the customer only after there is a legal agreement between the AA, the customer and the FIP; (3) to refrain from undertaking any business other than that of AA; (4) to refrain from taking the services of the third-party provider for undertaking the business of AA; and (5) to not store any financial data accessed on behalf of a customer.

The principal role of an AA will be to serve as a consent manager, ensuring that the data requested from the FIU is fetched from FIP and shared with FIU only after receiving the customer's consent.¹² Customers, through the AA, can restrict the consent in terms of the period for which data is shared with FIU and the types of data that need to be shared. The customer can at any time revoke their consent to share data with FIU. The AA maintains a log of consents provided and withdrawn. In other words, AA play the role of traffic cop in the data value chain by securely transmitting information from the FIP to the FIU and, in the process, facilitating easy sharing and consumption of data among various entities based on user consent.

Assessment of the last three requirements (3–5) for an entity to be an AA provides insight into an operating model for an AA. An AA cannot be an information broker; it

cannot process, enrich, cleanse or analyse data collected from the FIP. AA cannot employ a third party for data collection or screen scraping; it cannot store any financial information collected on behalf of a customer. These requirements ensure that AA earn their revenue from customers or FIU. AA cannot charge FIPs for collecting the data, nor can they sell customer data to third parties in any form. These requirements ensure that the profits of AA are capped and that AA is treated as a public utility.

Furthermore, to participate in the AA ecosystem, FIU needs to be an FIP, so it cannot hoard the data used to tailor its loan offering. FIU must abide by the principle of 'give and take' and, therefore, must assume the role of FIP to provide data requested by the user through an AA. AA ecosystem's institutional architecture ensures data portability and interoperability, preventing walling off data.¹³

The architecture allows the players in the AA ecosystem to collect data, but interoperability baked into the process prevents the data collectors from creating a monopoly on the acquired data. This institutional architecture is augmented by the technical architecture called 'Electronic Consent Framework', designed for consent sharing among the players in the AA ecosystem (also called AA, FIP and FIU).

How are data privacy, security and confidentiality maintained?

OCEN is a manifestation of DEPA (Data Empower and Protection Architecture), allowing the flow of data between FIP and FIU via APIs and the consent manager, also known as the AA.¹⁴ AA acts as a liaison between multiple data stewards and data users via a digital platform where the users can view their data fetched from FIPs and consent for the data to be shared with FIUs. An important feature of this architecture is the separation of data flow from consent collection. Second, consent to collect information from FIPs does

not necessarily imply consent to share the information. AA is data blind and does not store the data collected from FIPs but merely acts as a conduit. In addition, FIPs do not identify the end user of data.

Data portability and data interoperability are the building blocks of DEPA. Data portability enables the user or data principal to request the data holder (FIP) to download or transfer user data stored in FIP to another system. It ensures that user data, which is fundamentally non-rival data, is not siloed within FIPs. DEPA permits users to request portability from multiple sources simultaneously rather than individually. Data interoperability enables regular interaction and information exchange between systems through a standardised and well-documented interface. DEPA is a solution that integrates FIP and FIU via AA architecture, with the RBI taking the initiative to standardise APIs for data sharing. These open APIs can be plugged into the AA architecture to facilitate data sharing between FIPs and FIUs without needing point-to-point integration as depicted in Figure 2.

DEPA grants users complete control over the use of their data. A user registered with AA will be able to provide or revoke consent for sharing financial data with an FIP. In addition, the user determines the extent to which financial information can be shared with the FIU.¹⁵ For instance, if a user has two bank accounts and three investment accounts, he or she can choose only to provide information on two bank accounts and one investment account.

How is India's account aggregator model different from other models of open banking?

The Indian AA model is comparable to the UK's open banking model. In either case, data is shared between FIPs and FIUs via standardised APIs, as opposed to screen scraping, which requires users to provide login credentials to the data user. Data

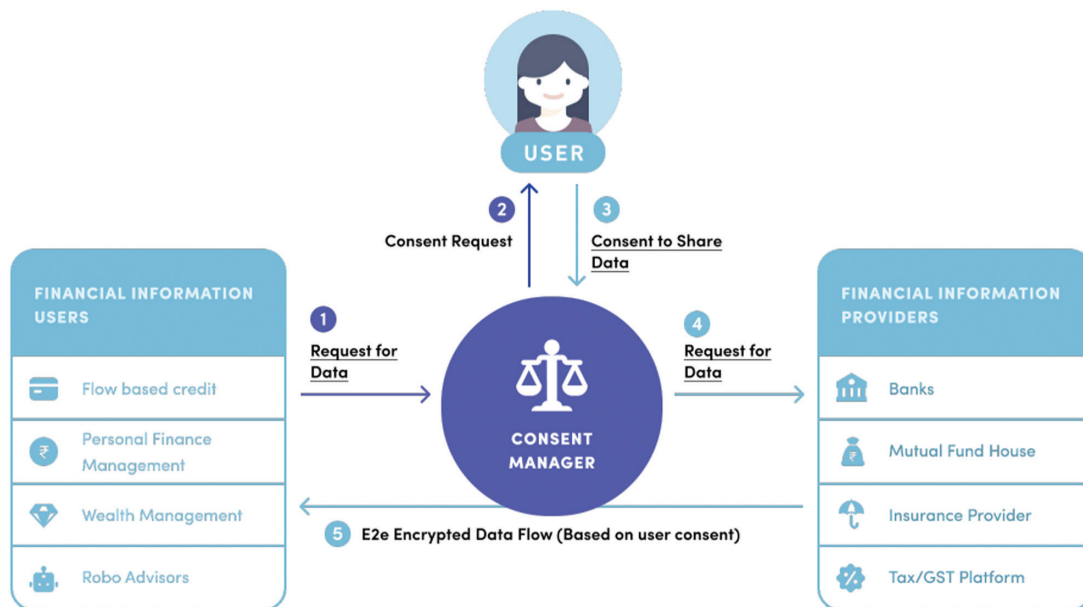


Figure 2 Data empowerment and protection architecture

Source: Roy, A., (August, 2020), 'Data empowerment and protection architecture', *Niti Aayog*, 56 pp., Figure 1, available at: <https://www.niti.gov.in/sites/default/files/2020-09/depa-book.pdf>

sharing occurs with the user's unequivocal consent and on their terms.

On the other hand, open banking in the UK and AA differ significantly in two important aspects. Unlike open banking, which provides licensed third-party AA with unfettered access to customer data, India's AA model covers regulated entities under RBI, SEBI, IRDA and PFRDA and could promote banking, insurance and investment activities. In contrast to India's AA, there are restrictions on the types of financial information shared in other nations through open banking.

Second, AA serves as the consent manager under the DEPA model.¹⁶ In contrast, in the UK, it is the responsibility of the data requestor to collect consent from the end user. In the AA model, FIPs and FIUs are linked via consent managers, and FIPs and FIUs are not required to have a point-to-point connection, as is the case in open banking. Additionally, the DEPA model distinguishes the entity (AA) that collects consent from the entity (FIU) that collects data. Individual consent is the responsibility of the information user in open

banking. Furthermore, the AA ecosystem ensures that the data is not stored with the AA but is a pass through from FIP to FIU, whereas the third-party AA in the UK can store, analyse and enrich the customer data.

What can be potential revenue model for account aggregator?

AA is a new class of NBFCs approved by RBI to manage consent for financial data sharing. Thus, creating a new player in the AA ecosystem adds a cost to the consumer of the AA service. The AA brings efficiency gains in the lending process that must offset the additional cost.

AA can charge the customers for managing their consent to transmit safely and securely from FIP to FIU. Customers may not, however, go to the AA directly to request data. Therefore, the AA process must be embedded in the loan boarding journey of FIUs so that customers benefit from FIUs' ability to provide better and more competitively priced loan products,

cross-sell the products and provide quicker turnaround time for credit decisioning. The customer and the FIU benefit from the data flow from FIP to FIU on customer consent. AA can charge either the customer or the FIU.

How data flows in the account aggregator ecosystem?

According to Datawani and Raman,¹⁷ the data flow is as follows:

1. An FIU initiates a request for customer information to an AA. The request includes the reason for the data request and the time frame for which the data is required (Figure 3).
2. AA send a request to the customer seeking permission to share their data for the stated purpose.
3. The client then responds. If the customer agrees to share their data, they must also provide account numbers and other pertinent information to obtain the requested data.
4. AA process the response received from the customer.
 - a. If consent is granted, the request is sent to the FIP to retrieve relevant

data. Using standard APIs, the data is retrieved.

- b. If consent is denied, the FIU is notified accordingly.
5. FIP receives the data request, examines it, retrieves the data from their systems, encrypts it and transmits it to the AA.
6. The AA collects data from one or more FIPs, aggregates the data and transmits the aggregated data to the FIU. AA are prohibited from storing, modifying or reading FIP data. AA, unlike credit bureaus, are data blind.

CURRENT STATISTICS

The RBI has currently accredited eight AAs. Approximately 23 banks serve as FIPs, and approximately 70 entities are live or live-enabled FIUs.¹⁸

The AA dashboard for FIPs currently has about 5m accounts linked, that is accounts linked by the account owner (Figure 4).

So far about 5.9m consents have been completed on the AA dashboard for users of financial information. AA has facilitated the process wherein the FIP has transmitted the requested data to the FIU (Figure 5).

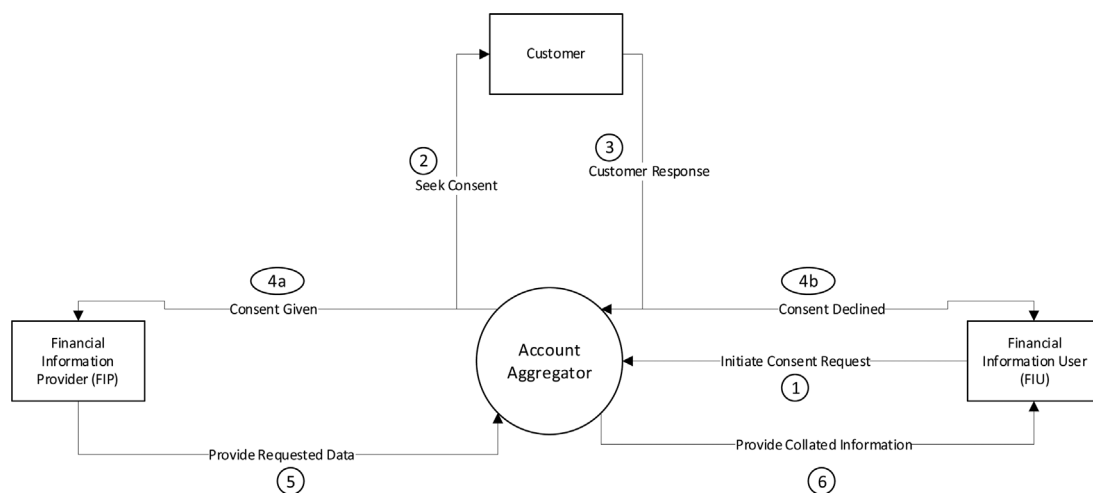


Figure 3 Account aggregator ecosystem

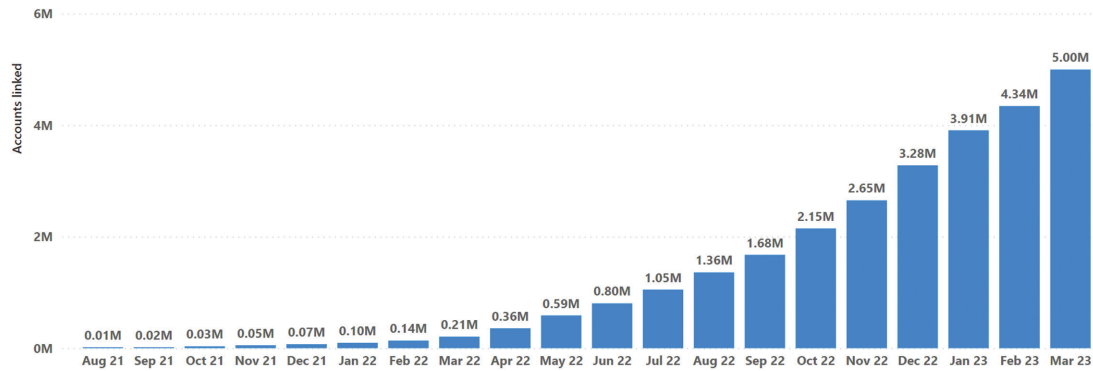


Figure 4 Count of accounts linked

Source: Sahamati, (19th March, 2023), 'Ecosystem dashboard', *DigiSahamati Foundation*, Figure 1, available at: <https://sahamati.org.in/aa-dashboard/>

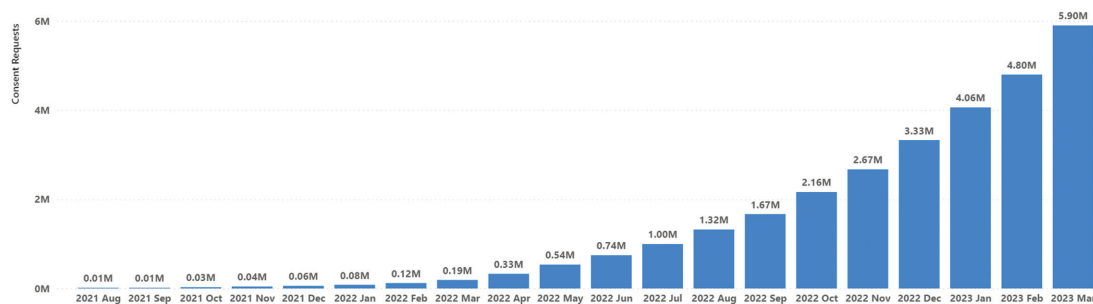


Figure 5 Count of consent request successfully fulfilled

Source: Sahamati, (19th March, 2023), 'Ecosystem dashboard', *DigiSahamati Foundation*, Figure 2, available at: <https://sahamati.org.in/aa-dashboard/>

USE CASES OF OCEN

In digital lending, there are two distinct use cases for AA.

Use case 1 is about streamlining the process of giving credit. India's current loan origination procedure has several obstacles. Bank statements are the primary input in the loan underwriting process, and the current protocol is to provide certified copies of bank statements or passbook images. Borrowers must visit multiple bank branches to collect physical copies of the statement. Additionally, some customers engage in statement fraud.¹⁹ Consequently, the process is fraught with inefficiencies, and there is always a risk of data privacy,

data recency and data trust, resulting in manual processes, longer turnaround times and high risk. An AA resolves these issues by facilitating the direct flow of data from FIPs to FIUs via the data-sharing rail without intermediaries.

Use case 2 is about keeping an eye on the loan and making any necessary changes that are of loan monitoring during the life of the loan.

Ease in extending credit

Borrowers who have financing need must search the credit market for an offer that meets their requirements.

A lender requires Know Your Customer (KYC) documentation and supporting financial statements such as bank statements, personal tax returns and GST documentation from the borrower to underwrite the loan. Using the e-KYC enabled by Aadhaar, that is identity rail, performs the borrower's KYC. FIU requests the necessary documentation for credit underwriting via the AA, which makes it easy for the borrower to share the financial data with a lender. A borrower can do all this without visiting various financial entities to get the documentation. They can provide all the documentation needed for securing a loan without having to step away from work in a safe, consented and secured manner using the AA.²⁰

A borrower must grant permission for the AA to share their data, which is recorded in the AA application. Using the consent, intent and duration for which data is required using standardised APIs, the AA collects and collates data from the FIP and provides it in real time or near real time to the FIU. FIU evaluates the information received from FIP via the AA and either approves or rejects the borrower's application. If the FIU approves the loan,

the borrower receives the funds via UPI. The loan is repaid through various channels, including UPI (Figure 6).

Loan monitoring

The bank must monitor the performance of a loan issued to a borrower to ensure that the loan remains collectible. If the borrower is an existing-to-bank (ETB) customer, then the bank can run analytics on the available data set to determine whether the loan will continue to perform. To assess the financial health of a borrower and to generate early warning signals, Banks must analyse statements of borrowers from other banks to comprehend the cash flows of the borrower, the number of days past due on active loans, and the new sales and buy date (GSTN).²¹ If there is no data rail, collecting these data signals is hard and can lead to mistakes.

Using the AA ecosystem, banks can monitor a borrower's financial health in real time or near real time and intervene if a loan is trending towards 'non-performance'. Banks can reduce their loan loss provisions as all loans are monitored in real time or close to real time. This use case allows

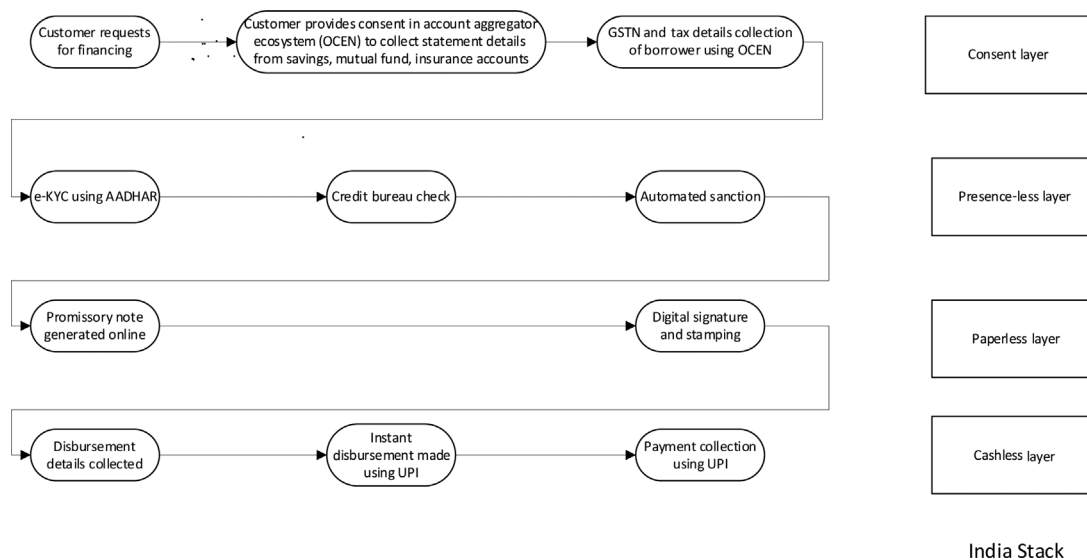


Figure 6 Data flow for loan underwriting

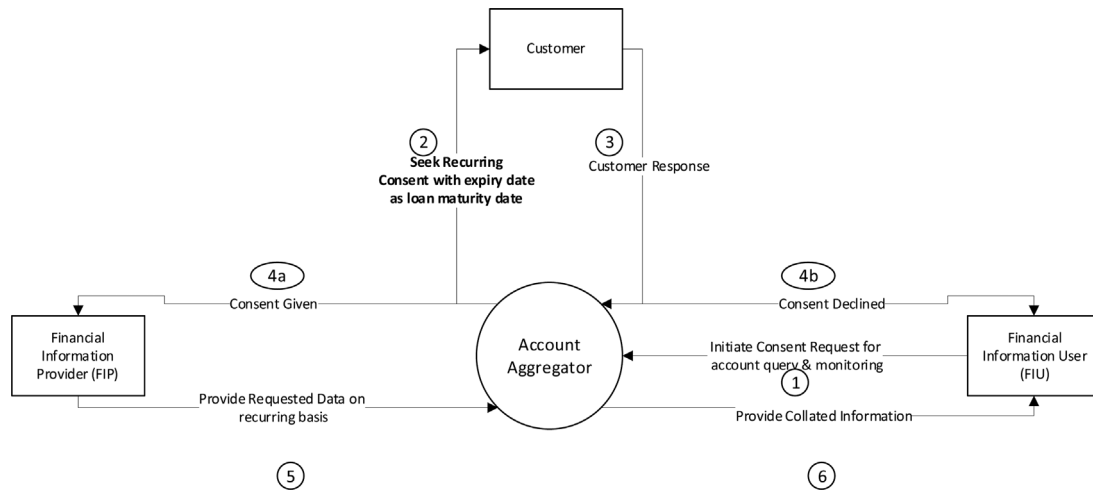


Figure 7 Data flow for loan monitoring

the bank to track cash flows for all loans, including business, personal and vehicle (Figure 7).

CONCLUSION

Digital technologies are connecting people like never before. Internet and smartphone penetration is making customers digital savvy; customers are using digital apps for fulfilling various needs such as grocery, ride-hailing apps, e-commerce and online streaming of movies. Consumers perform their banking activities online without stepping into brick-and-mortar branches. These activities generate vast digital footprints such as sales, purchase, banking transaction and tax information, creating information collateral. Information collateral can be used to assess credit risk. This data, however, is buried in legacy siloes. Digital interconnectedness has the potential to bring data from myriad sources on one platform and make effective use of the same. AA enabled by OCEN is one such endeavour of collecting and aggregating information from bank accounts, mutual funds holding and GSTN and providing the financial information to the requester of the data using standardised APIs in a safe and secured manner.

The number of adopters of the AA ecosystem is increasing with each passing day with new additions of FIPs and FIUs. Furthermore, the number of accounts linked on aggregator to fulfil data requests is also increasing with each passing day.

To expand the AA ecosystem's reach and faster adoption by end customers, FIUs and FIPs need to focus on user experience. FIU needs to integrate the AA framework's consent and other data handling aspects within their solution so that the customers do not have to go to multiple places to participate. Customers need to feel that it is one simple, smooth process from FIU to FIP, where consent is gathered from the customer with a few clicks and then data is fetched from FIPs. A customer has a frictionless and seamless induction experience. Furthermore, for financial inclusion and to enhance the reach to NTB and NTC, there needs to be an assisted AA framework with support in multiple languages; otherwise, only the creamy layer with smartphones will be able to participate in the AA ecosystem.

Second, customers should benefit from participating in the AA ecosystem. Customers will feel the perceived usefulness of the AA ecosystem only if there is faster credit decisioning on their loan application loan and real-time funds disbursal. A more

immediate credit decision is possible only when many FIPs are part of the ecosystem that can provide data points to the FIU in real time for faster credit decisioning. At this point, only the financial and tax entities, such as banks and GSTN, are participating in the ecosystem. Regulatory agencies must facilitate telecom companies' entry into the AA platform. Telecom companies based on the telecom services usage pattern can provide information on the consumer's consumption and income pattern, which are vital inputs for credit decisioning.

Data privacy, security and confidentiality are vital for the acceptability of the AA ecosystem. Any fraud or mishandling of the data on the platform would reduce the trust and attractiveness of the platform. It is therefore pertinent that the government of India bring in data privacy legislation sooner rather than later.

Financial inclusion or increasing access to finance requires two things. The first is to get unbanked within the formal financial network through digital financial infrastructure that is easy to use and has perceived usefulness. Second, there needs to be constant product innovation in providing financial services that leverage the digital financial infrastructure so that consumers do not leave the formal financial network. OCEN, which is provided as a public good, ensures the availability of trusted, high-quality financial data to lenders and, in the process, increases credit access to the NTB and NTC customer segments. Data-sharing rails provided by OCEN ensure that the data is not stuck in the silos but instead empowers the customers and businesses to benefit from their own data. Easier access to data with the customer's consent as recorded by AA allows financial entities to build innovative personalised financial products for the customers, fostering competition and efficiencies within the financial market. Nandan Nilekani recently said that the AA framework has the capability to democratise

credit so that millions of consumers without any credit history and millions of small businesses that could not get a loan earlier now have access to credit using information collateral.²²

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